JIEMING ZHANG

U.S. Citizen | 703-232-6926 | jiemingzhang23@gmail.com | github.com/jmyz23 | LinkedIn | Charlottesville, VA

EDUCATION

University of Virginia **Charlottesville, VA**

B.S. Computer Engineering & B.S. Electrical Engineering, ***GPA: 3.9*** *Expected May 2029*

Relevant Coursework: Computer Systems Organization I, Electric Circuits, Digital Logic Design, CPU Architecture, Data Structures & C++

TECHNICAL SKILLS

Languages:	C++, Python, VHDL, Java, Assembly
Hardware & EDA:	Custom PCB Design, KiCad, Altium, Quartus, ModelSim, ESP32, BNO055 IMU, MicroSD SPI, Multimeters
Protocols & Tools:	I2C, SPI, UART, CAN Bus, Git, Fusion 360
Concepts:	Embedded Firmware, Sensor Integration, Data Acquisition, Power Management, Signal Processing

ENGINEERING EXPERIENCE

Virginia Motorsports (Formula SAE) **Fall 2025 - Present**

Data Acquisition (DAQ) Subteam Member | University of Virginia

- Contributing to a redesign consolidating 5 discrete PCBs into one centralized sensor board handling 17 analog inputs across 4 sensor types via Teensy, reducing wiring complexity and failure points.
- Design schematics and multi-layer PCBs in Altium and KiCad for sensor integration and data logging on an all-electric FSAE racecar.
- Implement SD card logging from the shared DAQ struct, synchronizing readings from all 17 sensor channels to persistent storage.

TECHNICAL PROJECTS

IoT Golf Swing Analyzer & Visualizer Prototype **Summer 2026**

Embedded Hardware, Firmware & Python Analytics | Personal Project

- Designed and routed a custom ESP32 PCB (~$20 in components) with 3.7V LiPo-to-3.3V power management and interfaces for an IMU and MicroSD logger.
- Built non-blocking C++ firmware with button-controlled record/idle states to log 100 Hz BNO055 quaternion and accelerometer data over I2C to SPI MicroSD CSV storage.
- Created Python telemetry processing to correct gravitational drift, estimate clubhead velocity, and render 3D swing trajectories.

Automated Physical Sink Shutoff System **Fall 2025**

Introductory Engineering Design Project | UVA

- Engineered a retrofit shutoff mechanism for standard sink handles without internal plumbing access, integrating an ultrasonic sensor and calibrated servo-driven mechanical arm.

HACKATHON EXPERIENCE

VTHacks 14 **Sep 2026**

2-person team | Embedded Systems & Wireless Sensing

- Built an ESP32 occupancy-sensing prototype that scanned 2.4 GHz Wi-Fi and Bluetooth traffic for packet counts and RSSI.
- Developed C++ firmware to calculate a 0-100 density score from signal activity and fluctuation, calibrated against an empty-room baseline in nonvolatile memory.